

# UNIQUENESS OF THE STATIONARY LOGARITHMIC DLR STATE

**ABSTRACT.** We give an entropy comparison argument for uniqueness of the stationary one-dimensional logarithmic DLR state at every inverse temperature  $\beta > 0$ . The comparison theorem is first formulated under explicit count variance assumptions: a linear variance bound, sublinear variance growth, and integrability of the variance against  $t^{-2} dt$ . Two conditional laws with the same number of particles are compared with one deterministic lattice-boundary law carrying a weak quadratic confinement. A static concentration estimate for averages of finite gap vectors bounds their expectation difference by relative entropy per particle. Integrated discrepancy control makes the boundary contribution to the Fisher information tight, while sublinear variance makes the confinement cost negligible. We then derive the required variance assumptions from finite logarithmic electric-field energy by stationarization and a covariance spectral estimate. With the standard regularized interpretation of the renormalized energy, this identifies the stationary finite-energy solution of the canonical DLR equations with  $\text{Sine}_\beta$ .

*Manuscript status.* This is a working draft prepared for mathematical review. The argument is developed from the three supplied preprints listed in the bibliography. No external literature search or claim of priority is made. The versions actually used, including the energy convention discussed in Section 7, are specified explicitly.

## 1. INTRODUCTION AND STATEMENTS

The process  $\text{Sine}_\beta$  describes the microscopic bulk limit of one-dimensional logarithmic gases. Dereudre, Hardy, Leblé and Maïda [DHLM19] show that it satisfies canonical Dobrushin–Lanford–Ruelle equations and that every stationary finite-energy solution of those equations is number-rigid. Their Section 1.3 asks whether the stationary solution is unique. The conditional universality theorem of Kuijlaars and Miña-Díaz [KMD19] supplies a deterministic route at  $\beta = 2$ . For general  $\beta$ , Bourgade, Erdős and Yau [BEY12] develop a comparison method for local logarithmic Gibbs measures, together with strong particle-location estimates for convex analytic potentials.

The argument below uses the convexity and entropy ingredients of that method. It compares two candidate infinite-volume laws with the same finite reference law. It does not require a limiting theorem for the reference law, nor uniform microscopic rigidity of every frozen boundary. The finite-energy input is instead used to obtain an integrated discrepancy estimate.

We write  $\gamma$  for a locally finite simple counting measure on  $\mathbb{R}$ ,  $N_I(\gamma) = \gamma(I)$ , and  $\theta_a \gamma = \gamma - a$ . All stationary processes in the comparison theorem have intensity one. For such a process  $P$ , put

$$v_P(t) = \text{Var}_P N_{(0,t]}, \quad t > 0.$$

The canonical DLR property means that, conditionally on the exterior of a bounded interval and on its particle number, the interior points have the normalized logarithmic Gibbs density at inverse temperature  $\beta$ . The infinite exterior interaction is defined by symmetric cutoffs and relative energies, as in [DHLM19, Section 2.1].

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**Theorem 1.1** (Uniqueness under discrepancy control). *Fix  $\beta > 0$ . There is at most one stationary simple point process of intensity one satisfying the canonical logarithmic DLR equations and the following three conditions:*

$$v_P(t) \leq C_P t \quad (t > 0), \quad (1.1)$$

$$v_P(t) = o(t) \quad (t \rightarrow \infty), \quad (1.2)$$

$$\int_1^\infty v_P(t) t^{-2} dt < \infty. \quad (1.3)$$

The constant  $C_P$  need not be uniform over candidate processes.

For clarity, all three conditions are retained in the statement. Section 7 proves them together for each ergodic finite-energy DLR solution. The comparison proof itself requires no ergodicity.

**Theorem 1.2** (Finite-energy uniqueness). *Fix  $\beta > 0$  and interpret the renormalized logarithmic energy with the standard inner electric-field truncation specified in Section 7. If  $P$  is stationary, satisfies the canonical DLR equations of [DHLM19, Definition 3.1], and  $\mathbb{E}_P W < \infty$ , then*

$$P = \text{Sine}_\beta.$$

Theorem 1.2 follows from Theorem 1.1, the electric-field argument, and the ergodic DLR decomposition in [DHLM19, Proposition 3.13]. It makes no assertion about unnormalized conditional-density specifications allowing different intensities.

**1.1. Organization and main estimates.** Section 2 proves a static comparison inequality for averages of consecutive gap vectors:

$$|\mu(F_K) - \omega_{K,T}(F_K)|^2 \leq \frac{C_G T}{K} I(\mu \mid \omega_{K,T}).$$

Here  $\omega_{K,T}$  is a lattice-boundary logarithmic gas with quadratic confinement of strength  $T^{-1}$ . Section 3 collects the counting and Palm facts needed for random particle-ended intervals. Sections 4 and 5 control, respectively, the confinement cost and the exterior-force contribution. These estimates are assembled in Section 6. The electric-field argument in Section 7 verifies the hypotheses of Theorem 1.1 for finite-energy states.

## 2. A STATIC COMPARISON INEQUALITY FOR GAP STATISTICS

Fix  $K \geq 1$ , write  $L = K + 1$ , and let

$$\Omega_K = \{u \in \mathbb{R}^K : 0 < u_1 < \cdots < u_K < L\}.$$

Set  $u_0 = 0$ ,  $u_{K+1} = L$ , and  $g_i(u) = u_{i+1} - u_i$  for  $0 \leq i \leq K$ . The reference exterior is

$$\xi_K^{\text{lat}} = \{j \in \mathbb{Z} : j \leq 0 \text{ or } j \geq L\}.$$

With any fixed  $r_* \in (0, L)$ , define

$$H_K^{\text{lat}}(u) = - \sum_{1 \leq i < j \leq K} \log(u_j - u_i) - \sum_{i=1}^K \text{PV} \sum_{p \in \xi_K^{\text{lat}}} \log \left| \frac{u_i - p}{r_* - p} \right|. \quad (2.1)$$

The reference changes only an additive constant. Equivalently, Euler's product for the gamma function gives the completely finite expression

$$H_K^{\text{lat}}(u) = - \sum_{i < j} \log(u_j - u_i) + \sum_{i=1}^K \{\log \Gamma(u_i) + \log \Gamma(L - u_i)\} + \text{constant}.$$

The paired exterior series converges, and its derivatives of order at least two converge locally uniformly. The endpoint singularities are  $-\log u_i - \log(L - u_i)$ . For  $T > 0$  put

$$\omega_{K,T}(du) = Z_{K,T}^{-1} e^{-\Phi_{K,T}(u)} du, \quad \Phi_{K,T}(u) = \beta H_K^{\text{lat}}(u) + \frac{1}{2T} \sum_{i=1}^K (u_i - i)^2. \quad (2.2)$$

The normalizing constant is finite and strictly positive.

Fix an integer  $r \geq 1$  and a function  $G \in C_c^2((0, \infty)^r)$ . For  $K > r$ , define

$$F_K(u) = \frac{1}{K-r} \sum_{i=1}^{K-r} G(g_i(u), \dots, g_{i+r-1}(u)). \quad (2.3)$$

This uses only gaps between interior particles. Omitting finitely many edge windows has no effect on the limiting average.

For probability measures  $\mu \ll \omega$  on  $\Omega_K$ , our entropy and Fisher information conventions are

$$S(\mu \mid \omega) = \int \log \frac{d\mu}{d\omega} d\mu, \quad I(\mu \mid \omega) = \int \left| \nabla \log \frac{d\mu}{d\omega} \right|^2 d\mu.$$

Fisher information is interpreted in the weak Sobolev sense, with value  $+\infty$  outside its domain.

**Proposition 2.1** (Gap entropy bound). *For fixed  $\beta, r, G$ , there is a constant  $C_G$  independent of  $K, T$  such that, for  $K \geq 2r + 1$ ,*

$$|\mu(F_K) - \omega_{K,T}(F_K)|^2 \leq \frac{C_G}{K} S(\mu \mid \omega_{K,T}), \quad (2.4)$$

$$|\mu(F_K) - \omega_{K,T}(F_K)|^2 \leq \frac{C_G T}{K} I(\mu \mid \omega_{K,T}). \quad (2.5)$$

*Proof.* Write  $\Phi = \Phi_{K,T}$ . For  $v \in \mathbb{R}^K$ , extend  $v$  by  $v_0 = v_{K+1} = 0$ . Nearest-neighbor interactions and the two endpoint charges give

$$v^\top \Phi''(u) v \geq \beta \sum_{i=0}^K \frac{(v_{i+1} - v_i)^2}{g_i(u)^2} + \frac{1}{T} |v|^2. \quad (2.6)$$

The remaining logarithmic interactions have nonnegative Hessians.

Regard  $F_K$  as a function of the gaps and set  $q_i = \partial F_K / \partial g_i$ . Bounded overlap of the windows in (2.3) gives  $|q_i| \leq C_G/K$ . Moreover  $q_i = 0$  unless  $g_i$  lies in a fixed bounded interval determined by the support of the derivatives of  $G$ . Consequently,

$$\begin{aligned} |\nabla F_K \cdot v|^2 &= \left| \sum_i q_i (v_{i+1} - v_i) \right|^2 \\ &\leq \left( \beta^{-1} \sum_i q_i^2 g_i^2 \right) \left( \beta \sum_i \frac{(v_{i+1} - v_i)^2}{g_i^2} \right) \leq \frac{C_G}{K} v^\top \Phi'' v. \end{aligned}$$

Thus

$$\nabla F_K^\top (\Phi'')^{-1} \nabla F_K \leq C_G/K. \quad (2.7)$$

The same support property for second derivatives, followed by  $2|ab| \leq a^2 + b^2$  and bounded overlap, gives

$$|v^\top F_K'' v| \leq \frac{C_G}{K} v^\top (\beta H_K^{\text{lat}}) v. \quad (2.8)$$

Let  $\omega_s$  have density proportional to  $e^{sF_K} d\omega_{K,T}$ . For  $|s| \leq c_G K$ , (2.8) yields

$$(\Phi - sF_K)'' \geq \frac{1}{2} \beta H_K^{\text{lat}} + T^{-1} I \geq \frac{1}{2} \Phi''.$$

The Brascamp–Lieb variance inequality and (2.7) imply  $\text{Var}_{\omega_s} F_K \leq C_G/K$ . Integrating the second derivative of the logarithmic moment-generating function therefore gives

$$\log \mathbb{E}_{\omega_{K,T}} \exp\{s(F_K - \omega_{K,T}(F_K))\} \leq C_G s^2/K, \quad |s| \leq c_G K. \quad (2.9)$$

The entropy inequality, used with either sign, shows that for  $0 < s \leq c_G K$ ,

$$|\mu(F_K) - \omega_{K,T}(F_K)| \leq \frac{S(\mu | \omega_{K,T})}{s} + \frac{C_G s}{K}.$$

Optimizing proves (2.4) when the entropy is at most a fixed multiple of  $K$ . Above that threshold it follows from  $|F_K| \leq \|G\|_\infty$ , after enlarging the constant.

Finally, (2.6) gives the logarithmic Sobolev inequality

$$S(\mu | \omega_{K,T}) \leq \frac{T}{2} I(\mu | \omega_{K,T}), \quad (2.10)$$

and hence (2.5). The static convexity inequalities here are the ones used in [BEY12, Sections 3 and 5]. They apply on a convex simplex by approximation: first impose positive lower bounds on all gaps, use the convex-domain inequalities, and then let those bounds decrease to zero. The boundary terms have the favorable convex sign. Bounded tilts pass by dominated convergence, and the logarithmic Sobolev inequality passes first for smooth bounded functions and then by Sobolev truncation. This proof has no restriction  $\beta \geq 1$ .  $\square$

### 3. DISCREPANCY AND PALM PRELIMINARIES

Throughout Sections 3–6,  $P$  satisfies the hypotheses of Theorem 1.1. The symbol  $P^0$  denotes its Palm distribution, and  $T_*$  denotes the shift from a point at the origin to its successor.

**Lemma 3.1** (Almost-sure exterior regularity). *Almost surely the configuration has density one in both directions, and, at every finite origin  $a$ ,*

$$\int_1^\infty \frac{(N_{(a,a+t]} - t)^2 + (N_{[a-t,a)} - t)^2}{t^2} dt < \infty. \quad (3.1)$$

*The symmetric principal value  $\lim_{R \rightarrow \infty} \sum_{p \in \gamma, 0 < |p| \leq R} 1/p$  exists, with a possible point at the origin omitted. These properties hold  $P^0$ -almost surely as well.*

*Proof.* By (1.1), Chebyshev's inequality along  $t = n^2$  and Borel–Cantelli give  $N_{(0,t]}/t \rightarrow 1$ ; monotonic interpolation removes the subsequence. The backward statement is identical. Tonelli and (1.3) give (3.1) at  $a = 0$ . Write the oriented cumulative discrepancy as  $D_0(x)$  for  $x \in \mathbb{R}$ . The discrepancy of  $(a, a+t]$  equals  $D_0(a+t) - D_0(a)$ . A change of variable and comparability of  $t$  and  $a+t$  for large  $t$  prove the integrability at every finite  $a$ , simultaneously on the same event.

For the principal value, put  $D_{\text{sym}}(t) = N_{(0,t]} - N_{[-t,0)}$ . Then

$$\sum_{1 < |p| \leq R, p \in \gamma} \frac{1}{p} = \frac{D_{\text{sym}}(R)}{R} - D_{\text{sym}}(1) + \int_1^R \frac{D_{\text{sym}}(t)}{t^2} dt.$$

The first term vanishes by density one. The integral converges absolutely by Cauchy–Schwarz and (3.1). Add the finite sum over  $0 < |p| \leq 1$ . Since the resulting good event is invariant under all finite translations, Campbell's formula transfers it to Palm measure.  $\square$

**Lemma 3.2** (Integrable Palm count envelopes). *Under  $P^0$  define*

$$C^+ = \sup_{t \geq 0} \frac{N_{(0,t]}}{1+t}, \quad C^- = \sup_{t \geq 0} \frac{N_{[-t,0)}}{1+t}.$$

*Then  $\mathbb{E}_{P^0} C^+ + \mathbb{E}_{P^0} C^- < \infty$ .*

*Proof.* For a spatially rooted sample put  $C_0 = \sup_{t \geq 0} N_{(0,t]}/(1+t)$ . A dyadic bound gives

$$C_0 \leq N_{(0,1]} + 2 + \sum_{j \geq 0} 2^{-j} |N_{(0,2^{j+1}]} - 2^{j+1}|.$$

Minkowski's inequality and (1.1) show  $\mathbb{E}_P C_0^2 < \infty$ . For every  $p \in \gamma \cap (0, 1]$ ,  $C^+(\theta_p \gamma) \leq 2C_0(\gamma)$ . Campbell's formula at intensity one therefore gives

$$\mathbb{E}_{P^0} C^+ = \mathbb{E}_P \sum_{p \in \gamma \cap (0,1]} C^+(\theta_p \gamma) \leq 2\mathbb{E}_P [N_{(0,1]} C_0] < \infty.$$

Reflect the argument for  $C^-$ . □

**3.1. The first positive point and the preceding gap.** Under  $P^0$  enumerate the points by  $\dots < p_{-1} < p_0 = 0 < p_1 < \dots$ , and write  $g_i = p_{i+1} - p_i$ . Palm measure is invariant under  $T_*$ , and Campbell's identity gives the following version of the inspection paradox. If  $a$  is the first positive point of a spatial sample, the law of  $\theta_a \gamma$  is

$$\tilde{P}^0(d\gamma) = g_{-1}(\gamma) P^0(d\gamma). \quad (3.2)$$

Indeed, each gap ending at a point  $p$  contributes the interval of possible spatial origins of length  $g_{-1}(\theta_p \gamma)$ . In particular  $\mathbb{E}_{P^0} g_{-1} = 1$ .

The event that the entire gap containing the spatial origin has length at most  $M$  becomes  $\{g_{-1} \leq M\}$  in (3.2). On this event the density relative to  $P^0$  is at most  $M$ . Since  $T_*$  preserves  $P^0$ , the same domination holds after any fixed number of successor shifts, uniformly in that number. We will use this for the two endpoints of a block.

**Lemma 3.3** (Spatial block averages identify Palm gap laws). *For fixed  $r$  and bounded  $G$ , the expectation under  $P$  of the average of  $G$  over  $K - r$  successive gap windows following the first positive point converges to  $P^0(G(g_0, \dots, g_{r-1}))$ .*

*Proof.* Let  $\mathcal{I}$  be the invariant sigma-algebra of  $T_*$ . Birkhoff's theorem gives convergence of these bounded averages to  $\mathbb{E}_{P^0}[G \mid \mathcal{I}]$  under Palm measure. Density one and  $\mathbb{E}_{P^0} g_{-1} < \infty$  give

$$\mathbb{E}_{P^0}[g_{-1} \mid \mathcal{I}] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} g_{-1} \circ T_*^j = 1.$$

Use (3.2), dominated convergence with the integrable weight  $g_{-1}$ , and conditioning on  $\mathcal{I}$ . □

#### 4. DLR INTERVALS WITH A FIXED PARTICLE NUMBER

For a spatial sample let  $a$  be the first positive point and let  $b$  be the  $(K+2)$ -nd positive point. The open interval  $(a, b)$  contains exactly  $K$  points, written  $x_1 < \dots < x_K$ . Put

$$L = K + 1, \quad \ell = b - a, \quad c = L/\ell, \quad u_i = c(x_i - a).$$

Let  $\zeta = \gamma - \sum_{i=1}^K \delta_{x_i}$  denote the retained exterior. Both endpoints, all negative points, and all points beyond  $b$  are retained.

**Lemma 4.1** (Random-rank DLR disintegration). *Conditionally on  $\zeta$ , the  $x_i$  have the normalized logarithmic Gibbs density on  $a < x_1 < \dots < x_K < b$ , with exterior  $\zeta$ . After the affine change of variables the conditional law, denoted  $\mu_\zeta$ , is a logarithmic Gibbs law on  $\Omega_K$  with exterior  $c(\zeta - a)$  and endpoint charges at  $0, L$ .*

*Proof.* First restrict to  $b < R$  for a fixed integer  $R$  and disintegrate canonical DLR in  $(0, R)$ . Conditional on its exterior, particle number, and all ordered coordinates except the  $K$  middle points, the remaining coordinates have precisely the Gibbs density in the gap between  $a$  and  $b$ . The number in  $(0, R)$  is determined by the retained configuration and  $K$ . The selection of  $a$  as first positive point and of  $b$  as the specified rank is preserved by all such interior resamplings. Thus

the conditioning introduces no additional factor depending on the middle coordinates. The events  $\{b < R\}$  are retained-exterior measurable and exhaust probability one.

Equivalently, one may restrict the  $K$ -th Campbell density in [DHL19, Theorem 3.7] to the ordered tuple lying between the first two positive points of the reduced configuration. The affine Jacobian and the constant factors in logarithmic interactions are absorbed by normalization. Relative exterior energies are translation and scale covariant: a change of cutoff center changes only distant shells, whose contribution vanishes by density one and the relative summand bound  $O(1/|p|)$ .  $\square$

Write  $\mathfrak{g}_* = a - p_-$ , where  $p_-$  is the last point before the spatial origin. This is the whole gap containing that origin. It is measurable from  $\zeta$ , and  $a \leq \mathfrak{g}_*$ .

**Lemma 4.2** (Good endpoint lengths). *One has  $\ell - L = o_P(\sqrt{K})$ . For any two candidate processes  $P, Q$ , there is a deterministic sequence  $r_K \downarrow 0$  such that, for each fixed  $M$  and*

$$E_{K,M} = \{\mathfrak{g}_* \leq M, |\ell - L| \leq r_K \sqrt{K}\}, \quad (4.1)$$

$$\limsup_{K \rightarrow \infty} P(E_{K,M}^c) \leq P(\mathfrak{g}_* > M),$$

and the same assertion holds for  $Q$ . The event  $E_{K,M}$  is retained-exterior measurable.

*Proof.* For fixed  $\varepsilon > 0$ , the event that the  $(K+2)$ -nd positive point exceeds  $K+2+\varepsilon\sqrt{K}$  is a count deviation of at least  $\varepsilon\sqrt{K}$  at a deterministic interval of length asymptotic to  $K$ . Chebyshev and (1.2) make its probability tend to zero. The lower tail is identical. Since  $a$  is a finite random variable with a law not depending on  $K$ , subtracting  $a$  proves the assertion for  $\ell$ . Choose a common slowly decreasing  $r_K$  by diagonalization. The assertions about  $E_{K,M}$  follow.  $\square$

Define the conditional quadratic cost

$$J_K(\zeta) = \mathbb{E}_{\mu_\zeta} \sum_{i=1}^K (u_i - i)^2. \quad (4.2)$$

**Lemma 4.3** (Negligible quadratic cost). *For every fixed  $M$ ,*

$$K^{-2} \mathbb{E}_P[\mathbf{1}_{E_{K,M}} J_K] \rightarrow 0. \quad (4.3)$$

*Proof.* For deterministic ordered  $z_1, \dots, z_K \in (0, \ell)$ , let  $A(t)$  be their counting function and  $\rho = K/\ell$ . Direct expansion yields the exact identity

$$\int_0^\ell (A(t) - \rho t)^2 dt = \rho \sum_{i=1}^K \left( z_i - \frac{i - \frac{1}{2}}{\rho} \right)^2 + \frac{\ell}{12}. \quad (4.4)$$

For example,  $\int_0^\ell A(t)^2 dt = \sum_i (2i-1)(\ell - z_i)$  and  $\int_0^\ell tA(t) dt = \frac{1}{2} \sum_i (\ell^2 - z_i^2)$ , which verify (4.4).

Take  $z_i = x_i - a$  and put  $D_0(s) = N_{(0,s]} - s$ . Except at endpoints of Lebesgue measure zero,

$$A(t) - \rho t = D_0(a+t) + (a-1) + (1-\rho)t.$$

On  $E_{K,M}$ ,  $a \leq M$ ,  $\ell \asymp K$ , and  $b \leq 3K$  for all sufficiently large  $K$  (with  $M$  fixed). The midpoint locations  $(i - \frac{1}{2})\ell/K$  differ from  $i\ell/L$  by at most a fixed constant. It follows from (4.4) that, on this event,

$$\sum_i (u_i - i)^2 \leq C \int_0^{3K} D_0(s)^2 ds + C_M K + CK(\ell - L)^2. \quad (4.5)$$

Now  $\int_0^{3K} v_P(s) ds = o(K^2)$  by (1.2), and the last term on the good event is at most  $Cr_K^2 K^2$ . Take expectations. Because the good event is retained-exterior measurable, conditional expectation turns the left side into the expression in (4.3).  $\square$

## 5. TIGHTNESS OF THE EXTERIOR FORCE COST

The force estimates will only use tightness. We will not require an integrable product of an exterior energy and an interior count envelope.

**Lemma 5.1** (A half-line force envelope). *At a particle endpoint, remove the endpoint itself and count residual exterior points at outward distance at most  $t$ . For a density-one configuration satisfying Lemma 3.1, put*

$$D(t) = N_{\text{ext}}(t) - (t - 1)_+, \quad H(x) = \int_0^\infty \frac{|D(t)|}{(x+t)^2} dt, \quad x \geq 0.$$

Then

$$\mathcal{B} = H(0)^2 + \int_0^\infty H(x)^2 dx < \infty. \quad (5.1)$$

After an affine scale  $c \in [1/2, 2]$ , the analogous quantity centered at the new density  $\lambda = 1/c$  is bounded by  $C(\mathcal{B} + 1)$ .

*Proof.* Simplicity and local finiteness give a positive distance to the next residual exterior point. Thus  $D(t) = 0$  for all sufficiently small  $t$ . At infinity,  $\int D(t)^2 t^{-2} dt < \infty$  by Lemma 3.1. Cauchy–Schwarz proves  $H(0) < \infty$ . Applying the Schur test to the kernel  $t/(x+t)^2$ , with weight  $t^{-1/2}$ , gives

$$\int_0^\infty H(x)^2 dx \leq \frac{\pi^2}{4} \int_0^\infty D(t)^2 t^{-2} dt. \quad (5.2)$$

For the scaled configuration the centered discrepancy is

$$D(t/c) + (t/c - 1)_+ - c^{-1}(t - 1)_+.$$

The last two terms differ by a uniformly bounded function vanishing for  $t < 1/2$ . Its force envelope is at most  $C/(x + 1/2)$ . A change of variables in the first term proves the scaling assertion.  $\square$

Let  $B_\zeta(u)$  be the difference of the actual and lattice exterior forces in the rescaled block. More precisely, with endpoint terms cancelled,

$$B_\zeta(u) = \text{PV} \sum_{p \in c(\zeta - a) \setminus \{0, L\}} \frac{1}{u - p} - \text{PV} \sum_{p \in \xi_K^{\text{lat}} \setminus \{0, L\}} \frac{1}{u - p}.$$

For almost every retained exterior under a candidate law, each paired sum exists and their difference is bounded on  $[0, L]$ . Define

$$\mathcal{D}_K(\zeta) = \mathbb{E}_{\mu_\zeta} \sum_{i=1}^K B_\zeta(u_i)^2. \quad (5.3)$$

**Proposition 5.2** (Tight conditional force cost). *For fixed  $M$ , the nonnegative random variables  $\mathbf{1}_{E_{K,M}} \mathcal{D}_K$  are tight as  $K \rightarrow \infty$ .*

*Proof.* On the good event,  $c = L/\ell \in [1/2, 2]$  for large  $K$ . The scaled exterior density is  $\lambda = \ell/L$ . Center each scaled outward counting function at  $\lambda(t - 1)_+$ . Stieltjes integration by parts identifies its force difference from the corresponding continuum ray with an integral of  $D(t)/(u + t)^2$ . The boundary terms vanish by density one.

For the unit lattice the residual outward count is  $\lfloor t \rfloor$ ; its difference from  $(t - 1)_+$  is bounded and vanishes below 1. The continuum density mismatch between the two paired rays integrates explicitly to a signed version of  $(\lambda - 1) \log((u + 1)/(L - u + 1))$ . Consequently,

$$\begin{aligned} |B_\zeta(u)| &\leq H_+(u) + H_-(L - u) + \frac{C}{u + 1} + \frac{C}{L - u + 1} \\ &\quad + |\lambda - 1| \left| \log \frac{u + 1}{L - u + 1} \right|. \end{aligned} \quad (5.4)$$

The endpoint charges at  $0, L$  cancel exactly before this estimate. Here the half-line quantities  $\mathcal{B}_\pm$  depend only on the retained exterior.

For any decreasing nonnegative  $h$  and interior configuration, integration by parts gives

$$\sum_i h(u_i)^2 \leq C_{\text{int}}^+ \left( h(0)^2 + \int_0^L h(s)^2 ds \right), \quad C_{\text{int}}^+ = \sup_{t \geq 0} \frac{\#\{i : u_i \leq t\}}{1+t}. \quad (5.5)$$

For instance use  $N_{\text{int}}(t) \leq C_{\text{int}}^+(1+t)$  in  $\int h^2 dN_{\text{int}}$ ; the terminal boundary term gives exactly the expression on the right. The right endpoint has the reflected bound with  $C_{\text{int}}^-$ .

The function  $h(s) = \log((L+1)/(s+1))$  satisfies  $h(0)^2 + \int_0^L h(s)^2 ds \leq CL$ . Squaring (5.4), using (5.5), and taking conditional expectations gives

$$\mathcal{D}_K \leq C\{X_+(\mathcal{B}_+ + 1) + X_-(\mathcal{B}_- + 1)\} + C(\lambda - 1)^2 L(X_+ + X_-), \quad (5.6)$$

where  $X_\pm = \mathbb{E}_{\mu_\zeta} C_{\text{int}}^\pm$ .

It remains to verify tightness under the random boundary law. Lemma 5.1 gives  $\mathcal{B}_\pm < \infty$  Palm-almost surely. Equation (3.2), restricted to  $\mathfrak{g}_* \leq M$ , and Palm point-shift invariance bound the laws of the endpoint environments by  $MP^0$ , uniformly in  $K$ . The scaling bound in Lemma 5.1 therefore makes  $\mathcal{B}_\pm$  uniformly tight on this event.

The rescaled interior count envelopes are at most twice the corresponding full half-line envelopes of the unscaled configuration. Since  $E_{K,M}$  is exterior measurable, conditional expectation and Lemma 3.2 give

$$\mathbb{E}_P[\mathbf{1}_{E_{K,M}} X_\pm] \leq 2M\mathbb{E}_{P^0} C^\pm < \infty.$$

Thus  $\mathbf{1}_{E_{K,M}} X_\pm$  are tight as well. Products of tight nonnegative variables are tight, without any independence assumption. Finally,

$$(\lambda - 1)^2 L = \frac{(\ell - L)^2}{L} \leq Cr_K^2$$

on  $E_{K,M}$ . Insert these facts into (5.6).  $\square$

## 6. PROOF OF THE DISCREPANCY COMPARISON THEOREM

Let  $P, Q$  satisfy Theorem 1.1. Fix  $M, r, G$  and choose  $r_K$  in Lemma 4.2 common to the two processes. For almost every retained exterior, the logarithmic derivative of the density ratio  $\mu_\zeta/\omega_{K,T}$  has no contribution from the interior pair interactions or from the common endpoint charges. Therefore

$$I(\mu_\zeta \mid \omega_{K,T}) \leq C_\beta \mathcal{D}_K(\zeta) + \frac{2J_K(\zeta)}{T^2}. \quad (6.1)$$

For a fixed exterior, the residual exterior forces are bounded on the closed interval, since the next exterior particles have positive distance from the endpoints. Relative entropy and Fisher information are consequently finite. This also avoids an inverse-gap moment assumption for small  $\beta$ .

Define the deterministic sequence

$$q_K = \frac{1}{K^2} \left\{ \mathbb{E}_P[\mathbf{1}_{E_{K,M}} J_K] + \mathbb{E}_Q[\mathbf{1}_{E_{K,M}} J_K] \right\} \longrightarrow 0$$

and choose

$$\varepsilon_K = \max\{\sqrt{q_K}, K^{-1/2}\}, \quad T_K = \varepsilon_K K. \quad (6.2)$$

The same reference  $\omega_{K,T_K}$  is used for both processes. Proposition 2.1 and (6.1) imply

$$|\mu_\zeta(F_K) - \omega_{K,T_K}(F_K)|^2 \leq C_G \left\{ \varepsilon_K \mathcal{D}_K + \frac{J_K}{\varepsilon_K K^2} \right\}. \quad (6.3)$$

On the good event, the first term tends to zero in probability by Proposition 5.2; the second tends to zero in mean by Lemma 4.3 and (6.2). The left side is bounded by  $4\|G\|_\infty^2$ , so the expectation of its square root over good boundaries tends to zero.



Let  $\mathfrak{F}_K(\gamma)$  denote the rescaled block statistic  $F_K(u(\gamma))$ . Conditional DLR and exterior measurability of the good event imply

$$\limsup_{K \rightarrow \infty} |\mathbb{E}_P \mathfrak{F}_K - \omega_{K, T_K}(F_K)| \leq 2 \|G\|_\infty P(\mathfrak{g}_* > M).$$

The same estimate holds for  $Q$ . Cancelling the common reference and then sending  $M \rightarrow \infty$  proves

$$\lim_{K \rightarrow \infty} |\mathbb{E}_P \mathfrak{F}_K - \mathbb{E}_Q \mathfrak{F}_K| = 0. \quad (6.4)$$

The affine rescaling does not alter the limiting gap statistic. Extending  $G$  by zero gives a bounded globally Lipschitz function, and each gap occurs in at most  $r$  windows. Hence, with  $\mathfrak{F}_K^{\text{un}}$  the unscaled statistic,

$$|\mathfrak{F}_K - \mathfrak{F}_K^{\text{un}}| \leq \frac{C_G}{K-r} |c-1| \ell = \frac{C_G}{K-r} |\ell - L| \rightarrow 0$$

almost surely by density one. Boundedness gives convergence in mean. Lemma 3.3 and (6.4) now yield

$$\mathbb{E}_{P^0} G(g_0, \dots, g_{r-1}) = \mathbb{E}_{Q^0} G(g_0, \dots, g_{r-1})$$

for every  $r$  and  $G \in C_c^2((0, \infty)^r)$ . Thus all finite consecutive-gap distributions agree. Point-shift invariance extends this to every finite window of integer gap indices, so the Palm laws coincide. Palm inversion at intensity one,

$$\mathbb{E}_P f(\gamma) = \mathbb{E}_{P^0} \left[ \int_0^{g_0} f(\theta_t \gamma) dt \right],$$

then gives  $P = Q$ . This proves Theorem 1.1.

## 7. FROM LOGARITHMIC ELECTRIC ENERGY TO DISCREPANCY CONTROL

This section supplies the finite-energy implication separately from the comparison proof. We use the standard logarithmic regularization. Let  $\rho_\eta$  be uniform probability measure on the circle of radius  $\eta$  in  $\mathbb{R}^2$ . If  $-\operatorname{div} E = 2\pi(\gamma - dx \delta_0)$ , set

$$f_\eta(z) = \mathbf{1}_{\{|z| \leq \eta\}} \log(|z|/\eta), \quad F = E_\eta = E + \sum_{p \in \gamma} \nabla f_\eta(\cdot - (p, 0)).$$

Then

$$-\operatorname{div} F = 2\pi(\gamma * \rho_\eta - dx \delta_0). \quad (7.1)$$

*Remark 7.1* (Convention in the supplied version). The displayed cutoff in [DHLM19, Definition 2.20] is printed with  $|z| \geq \eta$ . That expression does not cancel the point singularities. Here the cutoff is explicitly taken with  $|z| \leq \eta$ , as required by the regularized electric-field interpretation and by (7.1). The argument below uses only finiteness of the spatial mean energy at one fixed  $\eta$ , so numerical normalizations of the renormalized-energy functional do not enter its conclusion.

**Lemma 7.2** (Stationary electric lift). *Let  $P$  be ergodic and suppose  $P$ -almost every configuration has finite renormalized electric energy. There are  $\eta > 0$  and a stationary joint law of  $(\gamma, F)$ , with configuration marginal  $P$ , satisfying (7.1) and*

$$\mathcal{E} = \mathbb{E} \int_{[0,1] \times \mathbb{R}} |F(x, y)|^2 dx dy < \infty. \quad (7.2)$$

*Proof.* Choose a  $P$ -generic configuration  $\gamma$  with finite energy. No attainment of the defining infimum over fields is needed: choose a compatible field with finite renormalized functional. For at least one sufficiently small fixed  $\eta$ , this gives

$$A = \limsup_{R \rightarrow \infty} \frac{1}{2R} \int_{[-R, R] \times \mathbb{R}} |E_\eta|^2 < \infty.$$

Write  $F = E_\eta$  and average translates of the single pair:

$$Q_R = \frac{1}{2R} \int_{-R}^R \delta_{(\theta_x \gamma, \theta_x F)} dx.$$

Its configuration marginal converges to  $P$ . For each  $a, b > 0$ ,

$$\mathbb{E}_{Q_R} \int_{[-a,a] \times [-b,b]} |F|^2 \leq \frac{2a}{2R} \int_{[-R-a, R+a] \times \mathbb{R}} |F|^2.$$

Local  $L^2$  bounds give tightness in a local distribution topology, for instance local negative Sobolev spaces where bounded  $L^2$  sets are relatively compact. Together with configuration tightness, a diagonal subsequence has a joint limit. Translation invariance follows because translating an averaging interval by a fixed amount changes its normalized mass by  $O(R^{-1})$ . The divergence identity passes in distributions: smearing the configuration by the fixed compactly supported circle measure is continuous in this topology. Lower semicontinuity on bounded vertical strips, followed by monotone convergence, proves (7.2). Thus no measurable configuration-by-configuration selection of fields is required.  $\square$

**Proposition 7.3** (Spectral energy and hyperuniformity). *Suppose  $P$  is stationary of intensity one, satisfies (1.1), and admits a stationary electric lift as in Lemma 7.2. Then (1.2) and (1.3) hold.*

*Proof.* Let  $\sigma = \gamma - dx$ . Finite local second moments define its covariance spectral measure  $S$ , normalized by

$$\mathbb{E}_P |\langle \sigma, f \rangle|^2 = \int_{\mathbb{R}} |\hat{f}(k)|^2 S(dk), \quad \hat{f}(k) = \int_{\mathbb{R}} e^{-ikx} f(x) dx.$$

The spectral measure can be singular. For interval counts,

$$v_P(t) = \int_{\mathbb{R}} \frac{4 \sin^2(kt/2)}{k^2} S(dk), \quad (7.3)$$

with the continuous value  $t^2$  at zero. The linear bound implies  $S(\{0\}) = 0$ .

Choose a real even smooth  $\chi$  with compact spectral support away from zero. Let  $\varphi(x, y)$  be the stationary random potential obtained by convolving  $\sigma$  horizontally with multiplier  $\chi(k)e^{-|k||y|}$ . Its mean is zero. Direct spectral calculation gives the averaged gradient energy

$$\mathbb{E} \int_{[0,1] \times \mathbb{R}} |\nabla \varphi|^2 = 2 \int_{\mathbb{R}} |k| \chi(k)^2 S(dk). \quad (7.4)$$

Pair (7.1) with this potential, average per unit horizontal length, and take expectation. The result is

$$\mathbb{E} \int_{[0,1] \times \mathbb{R}} F \cdot \nabla \varphi = 2\pi \int_{\mathbb{R}} \chi(k) a_\eta(k) S(dk), \quad (7.5)$$

where

$$a_\eta(k) = \frac{1}{2\pi} \int_0^{2\pi} e^{ik\eta \cos \theta - |k|\eta |\sin \theta|} d\theta.$$

For completeness, (7.5) is a stationary integration-by-parts identity, not a pointwise evaluation of an  $L^2$  field. First multiply the test by smooth horizontal and vertical cutoffs and smooth its cusp at  $y = 0$ . The distribution identity then holds pathwise, even though the test is a function of  $\gamma$ . After division by horizontal length, the horizontal cutoff terms vanish by stationarity and Cauchy–Schwarz. The spectral support bounded away from zero gives exponential vertical decay and finite second moments for the test and its gradient. It therefore permits removal of the vertical cutoff and the smoothing. The deterministic background term, including the deterministic difference between the smeared line and the original line, has zero expected pairing because the test has mean zero. The remaining covariance pairing is the multiplier on the right of (7.5).

Equations (7.2)–(7.5) and Cauchy–Schwarz imply, with a numerical constant  $C$ ,

$$\left| \int \chi a_\eta dS \right|^2 \leq C\mathcal{E} \int |k| \chi^2 dS. \quad (7.6)$$

The function  $a_\eta$  is real, continuous and equals one at zero. Choose  $\delta > 0$  such that  $a_\eta \geq 1/2$  on  $|k| < \delta$ . For a smooth even cutoff  $0 \leq \psi \leq 1$  supported in  $\varepsilon < |k| < \delta$ , take  $\chi = a_\eta \psi / |k|$ . Smooth approximation within this annulus is equivalent if needed. Writing

$$A_\psi = \int \frac{a_\eta(k)^2}{|k|} \psi(k) S(dk),$$

the right-hand integral in (7.6) is at most  $A_\psi$ , since  $\psi^2 \leq \psi$ . Thus  $A_\psi \leq C\mathcal{E}$ . Exhausting the punctured interval proves

$$\int_{0 < |k| < \delta} \frac{S(dk)}{|k|} < \infty. \quad (7.7)$$

This cutoff argument does not assume the desired integrability.

The high-frequency estimate follows from  $\int_1^2 v_P(t) dt < \infty$  and (7.3). For fixed  $\delta > 0$ , the continuous function  $\int_1^2 \sin^2(kt/2) dt$  has a strictly positive lower bound on  $|k| \geq \delta$ : it is positive on every compact annulus and tends to  $1/2$  as  $|k| \rightarrow \infty$ . Hence

$$\int_{|k| \geq \delta} k^{-2} S(dk) < \infty. \quad (7.8)$$

Tonelli in (7.3), followed by the elementary bound

$$\frac{4}{k^2} \int_1^\infty \frac{\sin^2(kt/2)}{t^2} dt \leq \begin{cases} C/|k|, & 0 < |k| < \delta, \\ 4/k^2, & |k| \geq \delta, \end{cases}$$

proves (1.3). Finally, the integrand in  $v_P(t)/t$  tends to zero for every  $k \neq 0$  and has the same two integrable majorants for  $t \geq 1$ . Dominated convergence proves (1.2).  $\square$

*Proof of Theorem 1.2.* Lemma 2.23 of [DHL19] gives

$$\mathbb{E}_P(N_I - |I|)^2 \leq C(C + \mathbb{E}_P W)|I|.$$

Stationarity then forces intensity one: otherwise the square of the mean discrepancy would grow quadratically in  $|I|$ . The same estimate gives (1.1). Proposition 3.13 of [DHL19] decomposes  $P$  into ergodic stationary canonical DLR solutions with finite expected energy. For each component, Lemma 7.2 and Proposition 7.3 give the remaining conditions of Theorem 1.1. All such components therefore coincide.

The existence theorem of [DHL19] supplies the stationary finite-energy solution  $\text{Sine}_\beta$ . Apply the same decomposition to it. Its components belong to the same singleton class, so their common law is  $\text{Sine}_\beta$ . Every candidate  $P$ , being a mixture of that law, equals  $\text{Sine}_\beta$ .  $\square$

## 8. RELATION TO CONDITIONAL UNIVERSALITY

At  $\beta = 2$ , the deterministic theorem of [KMD19, Theorem 1.3] provides another route. The reciprocal principal value and density-one conditions needed there follow from the discrepancy bounds, so its kernel convergence applies to every admissible candidate boundary. DLR and convergence of compactly supported Laplace functionals then identify the process.

For general  $\beta$ , the main theorem of the supplied version of [BEY12] concerns a fixed real analytic uniformly convex potential on the line, and its local comparison is combined with strong particle-location estimates. A direct application to arbitrary random DLR exterior potentials would require checking assumptions not provided by macroscopic potential convergence alone. The present proof uses the static convexity inequalities from that framework and compares with one common reference

law. In particular, it establishes neither a separate universality theorem for all deterministic boundaries nor a limiting theorem for  $\omega_{K,T_K}$ .

The distinction between the two energy inputs is useful:  $v_P(t) = o(t)$  makes the average quadratic displacement  $o(K)$  per particle, whereas  $\int_1^\infty v_P(t)t^{-2} dt < \infty$  supplies the half-line force envelopes. The linear bound alone gives neither of these two improvements. The weak confinement can be chosen between the quadratic displacement scale and the block length precisely because the improvement is little- $o$ ; no fixed power rate is needed.

## REFERENCES

- [BEY12] Paul Bourgade, László Erdős and Horng-Tzer Yau, *Universality of General  $\beta$ -Ensembles*. Supplied preprint arXiv:1104.2272v5, 5 February 2012. All theorem and equation references in this manuscript refer to that version. The bibliography of [DHLM19] records the published article in *Duke Mathematical Journal* **163** (2014), no. 6, 1127–1190.
- [DHLM19] David Dereudre, Adrien Hardy, Thomas Leblé and Mylène Maïda, *DLR equations and rigidity for the Sine-beta process*. Supplied preprint arXiv:1809.03989v2, arXiv version date 15 November 2019; title-page date 18 November 2019.
- [KMD19] Arno B. J. Kuijlaars and Erwin Miña-Díaz, *Universality for conditional measures of the sine point process*. *Journal of Approximation Theory* **243** (2019), 1–24. Supplied preprint arXiv:1703.02349v2, 21 March 2019.